M.Tech Part I/PhD Mining Engineering
Mid-Term Examination, 2025
MN573: Safety & Risk Management in Mines

Time: 2 Hrs. **Marks: 20**

1. Discuss how Heinrich's Safety Philosophy is a phenomenal departure from the then prevailing safety concept. (4)
   Using the concept of 'Domino Game' elaborate the injury sequence. What was his emphasis on prevention of accidents at workplaces?

2. Explain the process of risk management. Explain the process of risk assessment using qualitative methods developing a risk matrix. (4)

3. Explain various indicators used for measuring safety performance in mines and other industrial sectors with their relative merits and demerits. (4)

4a. What is a random variable? Can the sample mean ever be exactly equal to the population mean?–Under what circumstances? (2)

4b. Derive an expression for the standard deviation of a binomial random variable with parameters n and p. (2)

5a. A probability density function (PDF) is given as: (2)

$$f(x) = \begin{cases} \frac{3}{8}x^2; & 0 \leq n \leq 2 \\ 0; & otherwise \end{cases}$$

(i) Verify that this is a valid PDF.

(ii) Find P $(1 \leq X \leq 2)$.

5b. (i) What do you understand by safety? Is there any relationship between safety and risk. (1)

(ii) In an underground coal mine, the accident statistics for the year 2023 revealed that there were two fatalities, 5 serious bodily injuries and 14 reportable injuries. The coal mine operates 310 days in a year, three shifts in a day and eight hours in a shift. The average employment in a shift was 400 and the production of the mine was 0.85 million tons. Calculate serious bodily injury rate per million tons output. (1)

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